Experiment No. 22 (Mixture No. 3)

Date: / /

Aim : Analyse two acidic (anion) radicals qualitatively from given inorganic mixture.

Apparatus : Test tubes, test tube holder, test tube stand, filter paper etc.

A. Preliminary tests:

Test	Observation	Inference
Color	white / colourless	Chloride, nitrate, sulphate, carbonate of Ca^{2+}, Ba^{2+}, Br^{2+}, or Mg^{2+} may be present.
Nature	Crystalline (Soluble in water)	Cl^-, Br^-, I^-, NO_2^-, SO_4^-, CO_3^{2-}, CH_3COO^- or compounds of NH_4^+ or K^+ may be Present

B. Dry Tests for Acidic radicals (Test tube must be dry for following tests)

Test	Observations	Inference
1. Heating in a dry test tube:- Take a small quantity of the mixture in a clean and dry test tube and heat it strongly	Colourless, odurless gas which turns lime water milky.	CO_3^{2-}, $C_2O_4^-$ may be Present
2. Action of dil. H_2SO_4 : Take a small quantity of mixture + dilute H_2SO_4	Evolution of colourless odurless gas with brisk effervescence of CO_2 gas which turns freshly prepared lime water milky	CO_3^{2-} may be Present
3. Action of conc. H_2SO_4 : Take a small quantity of mixture + conc. H_2SO_4	yellowish brown gas having Pungent smell turns starch Paper yellow	Br^- may be Present
4. Action of Cu foils and conc. H_2SO_4 Mixture + Cu filings and conc. H_2SO_4, heat it strongly	—	—
5. Action of MnO_2 and conc. H_2SO_4 Mixture + MnO_2 powder + conc. H_2SO_4	Reddish brown gas during starch paper yellowish brown	Br^- may be Present.

Individual dry tests for SO_4^{2-} & PO_4^{3-} (if required)

1. Test for sulphate Mixture + dil. HCl, boil the solution + $BaCl_2$ solution	—	—
2. Test for phosphate Mixture + conc. HNO_3 boil + excess of ammonium molybdate solution	—	—

C. Preparation of original solution (O.S) : Mixture is dissolved in 20 mL quantity of distilled water in a beaker, stir with glass rod to dissolve the mixture. Clear solution is obtained. Use this O.S. for further tests of acidic radicals.

D. Wet Tests for Anion (Acidic Radicals)

1. O.S. + $AgNO_3$	white ppt insoluble in dil HNO_3 with effervescence of CO_2	CO_3^{2-} may be Present
If white ppt obtained in above test, perform following distinction test		
Distinction between Cl^-, Br^- and I^- : O.S. + dil. H_2SO_4 (till acidic) + chloroform + Cl_2 water (fresh) in excess, shake vigorously and observe the colour of chloroform layer carefully.	Chloroform layer yellowish brown	Br^- present
2. O.S. + $Ba(NO_3)_2$ solution	white ppt double in dil HCl or H_2SO_4 with efferrescence	CO_3^{2-} present
3.O.S. + dil. acetic acid + (Freshly prepared) $FeSO_4$ solution		
4. O.S. + diphenyl amine + conc. H_2SO_4		
5. O.S. + $FeCl_3$ solution		
6. O.S. + dil. acetic acid + $CaCl_2$		
7. O.S. + dil. H_2SO_4 + 2-3 drops of $KMnO_4$		

E. Confirmatory tests for Acidic Radicals

1. C.T. For first detected acidic radical ..CO_3^{--} (carbonate)

Test	Observation	Inference
1. O.S. + 2-3 drops of phenophtheline indicator	Pink colour	CO_3^{2-} confirmed
2. O.S + $Ba(NO_3)_2$	white ppt soluble in dil HNO_3	CO_3^{2-} confirmed

2. C.T. For second detected acidic radical ....

Test	Observation	Inference
1. O.S + MnO_2 + conc. H_2SO_4 and heat	Reddish brown gas turing starch paper yellowish brown	Br^{2-} confirmed
2. O.S + Cl_2 water + $CHCl_3$ shake well	chloroform layer acquire yellow colour.	Br^- confirmed

Result :

The given inorganic mixture no. 3 contains following two anions (Acidic radicals)

i) CO_3^- (Name of anion) (carbonate)

ii) Br .... (Name of anion) (Bromide)

Remark and sign of teacher: ....

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MCQ

Select the correct answer for the following multiple choice type questions.

1. A colourless solid 'A' produces black spots on the skin. It's aqueous solution gives brown ring test and also gives yellow ppt. with KI solution 'A' could be
   a. copper nitrate
   b. zinc nitrate
   c. silver nitrate
   d. lead nitrate
2. Which of the following sulphide is completely precipitated in acidic medium
   a. HgS b. PbS c. CdS d. CuS
3. The colour of precipitate observed, when potassium iodide is added to water soluble lead salts is
   a. black b. white c. yellow d. red
4. Which of the following is NOT a preliminary test
   a. charcoal cavity test
   b. flame test
   c. NaOH test
   d. brown ring test
5. The colour of carbonate cation precipitate of group V is.....
   a. black
   b. green
   c. white
   d. yellow

Short answer questions

1. **The removal of H_2S gas from filtrate is must before doing analysis of group III and V. Why?**

Ans. Due to Common ion effect ionisation of H_2S decreases less Sulphide ions are obtained these Sulphide ions suffient for the prepirctation of Second go cations in the form of them solubili prod

2. **Write the name of brown ppt. obtained, when Nessler's reagent is added in O.S. for confirmatory test of ammonium ion.**

Ans. Mercury (II) amido iodine is the name Brown ppt which is obtained when Nessler's reagent is added in O.S

3. **During the detection of groups, if Group-I is detected then for next Group test what care should be taken?**

Ans. The filtrate which is obtained from Group-I then carefully used for the detection of upcoming group

4. **Why cracking noise/decrepitation is observed during dry test in qualitative analysis of cations?**

Ans. when certain chemical compounde are heated it refers to the cracking or breaking up of less of limestone during (such compound) include lead Nitrate and airline

5. **Name a cation, which is not obtained from metal.**

Ans. $NH_4^{\oplus}$ Ammonium

Remark and sign of teacher: ....